A Generalized Bulk Parameterization Approach for Land-Atmosphere Heat Fluxes

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Bulk aerodynamic method

$$\overline{w'\theta'} = C_h |U| (\theta_s - \theta)$$

$$\overline{w'q'} = C_q |U| (q_s - q)$$

$$u_*^2 = C_m |U|^2$$

Stull 1993; Taylor 1916

Revised bulk parameterisation for heat fluxes

$$\overline{w'T'} = C\sigma_w(T_s - T_{air})$$

$$\overline{w'q'} = C\sigma_w(q_s - q_{air})$$

Revised bulk parameterisation for heat fluxes

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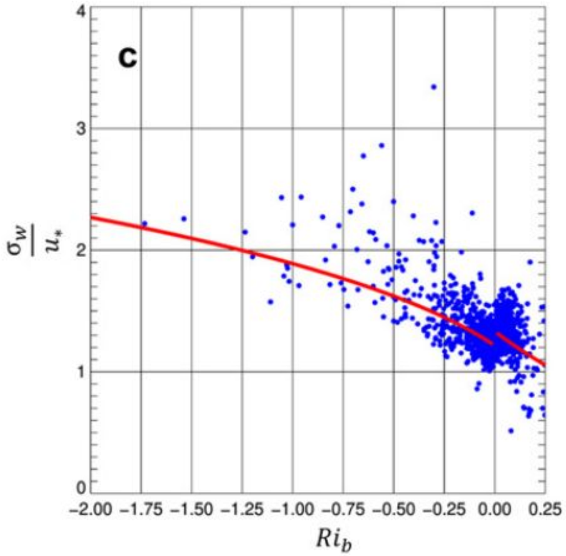


Unstable, Rib < 0

$$\frac{\sigma_{u,v,w}}{u_*} = \lambda_{\sigma_{u,v,w}}(1 - \omega_{\sigma_{u,v,w}}\mathrm{Ri}_b)^{1/3}$$

Stable, Rib > 0

$$\frac{\sigma_{u,v,w}}{u_*} = \chi_{\sigma_{u,v,w}}\exp(\gamma_{\sigma_{u,v,w}}\mathrm{Ri}_b)$$

$$Ri_b = \frac{g\Delta\theta_v\Delta z}{\overline{\theta_v}(\Delta\overline{u}^2\Delta\overline{v}^2)}$$

Lee and Meyers 2023

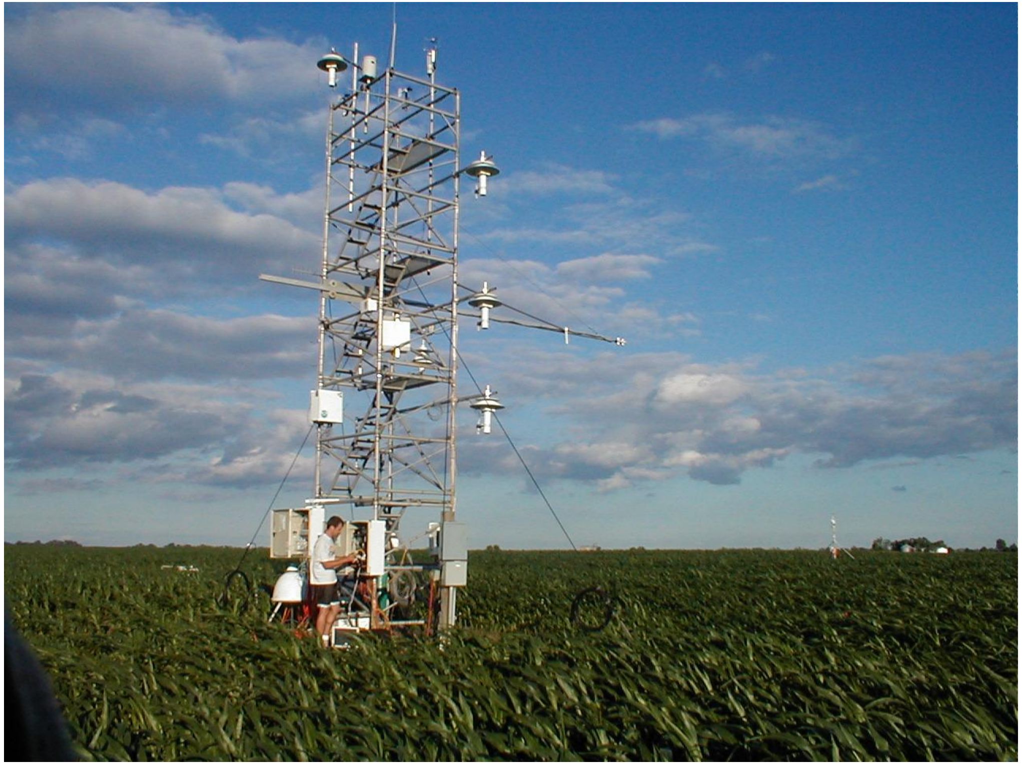
Revised bulk parameterisation for heat fluxes

$$\frac{\sigma_w}{U} = f(Ri_b)$$

$$\overline{w'T'} = C\sigma_w(T_s - T_{air}) = CUf(Ri_b)(T_s - T_{air})$$

$$\overline{w'q'} = C\sigma_w(q_s - q_{air}) = CUf(Ri_b)(q_s - q_{air})$$

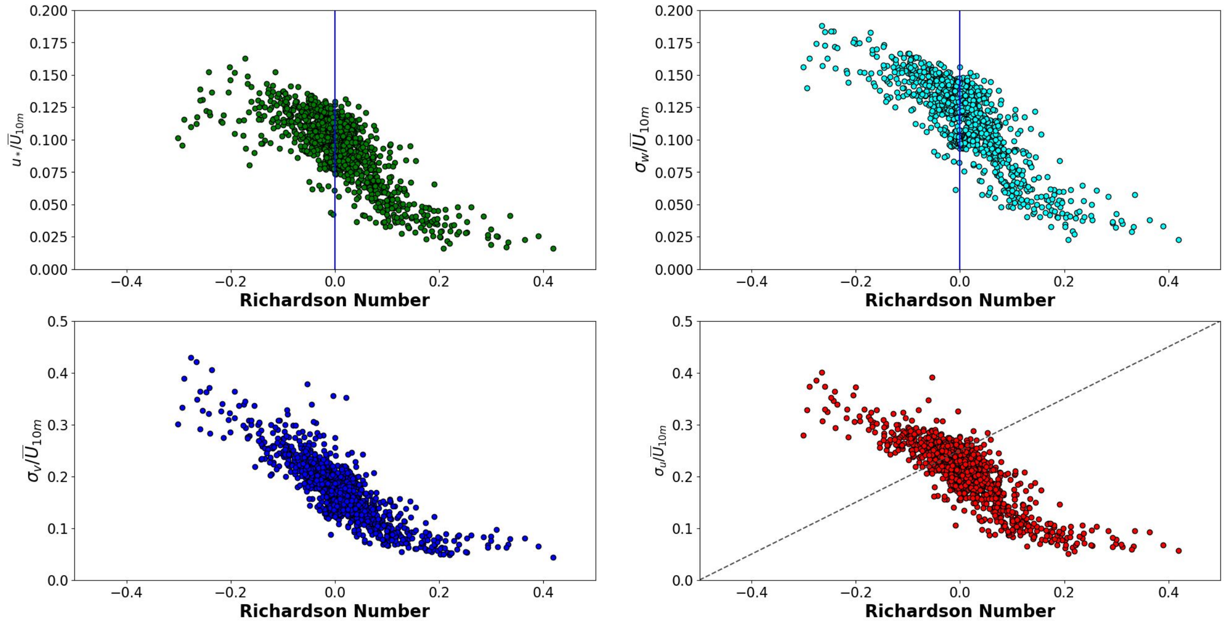
ARL Surface FLux Station, Bondville, IL



Installed June 2001

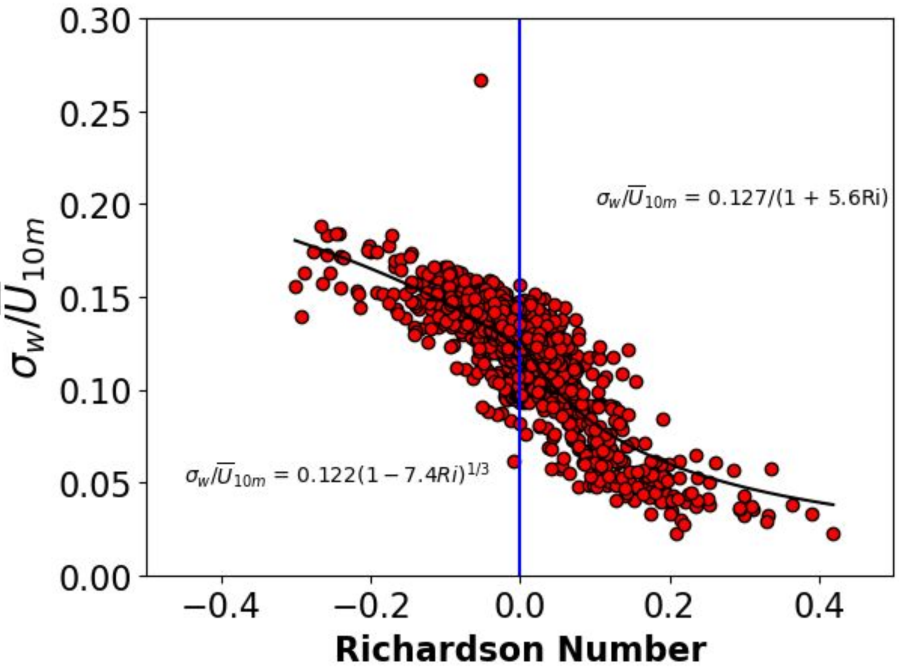
- 3 level wind and Ta profiles
- 4-component radiometer
- Uniform Soybean/corn canopy

Turbulence Statistics vs Stability (Jul - Aug 2013)



$$Ri_b = \frac{g\Delta T\Delta z}{\overline{T}(\Delta\overline{u}^2\Delta\overline{v}^2)}$$

Scaling relationship



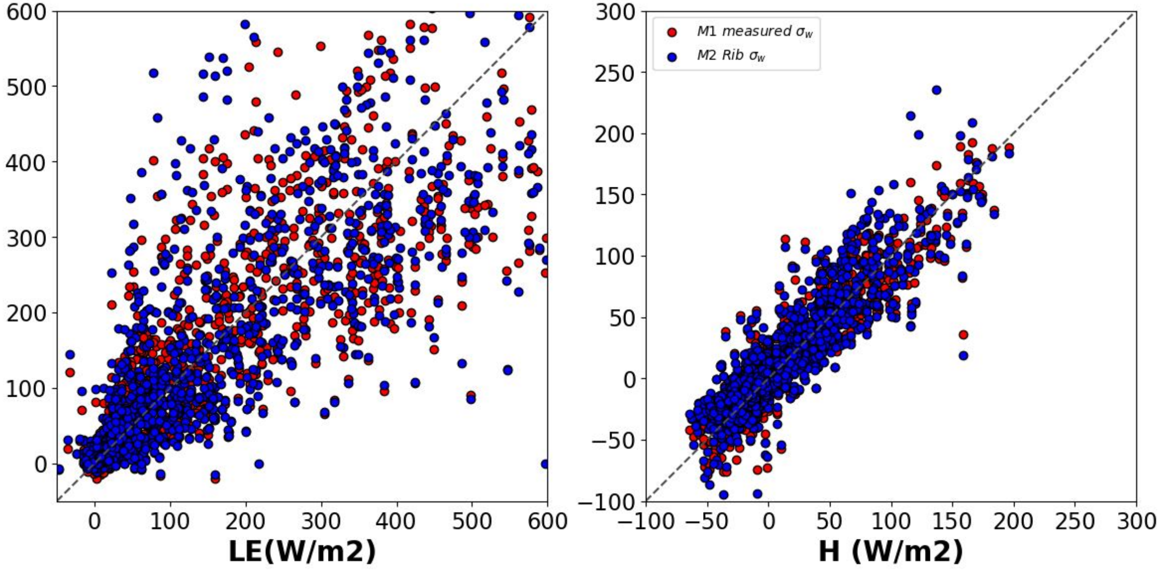
unstable

$$\sigma_w/\overline{U}_{10m} = 0.122(1 - 7.4Ri)^{1/3}$$

stable

$$\sigma_w/\overline{U}_{10m} = 0.127(1 + 5.6\mathrm{Ri})$$

Fully parameterized fluxes BNV Jul - Aug 2013



M1 M2

$$\overline{w'T'} = C\sigma_w(T_s - T_{air}) = CUf(Ri_b)(T_s - T_{air})$$

$$\overline{w'q'} = C\sigma_w(q_s - q_{air}) = CUf(Ri_b)(q_s - q_{air})$$

How well does this hold up for complex sites?

ARL Surface FLux Station, Crested Butte, CO

SPLASH

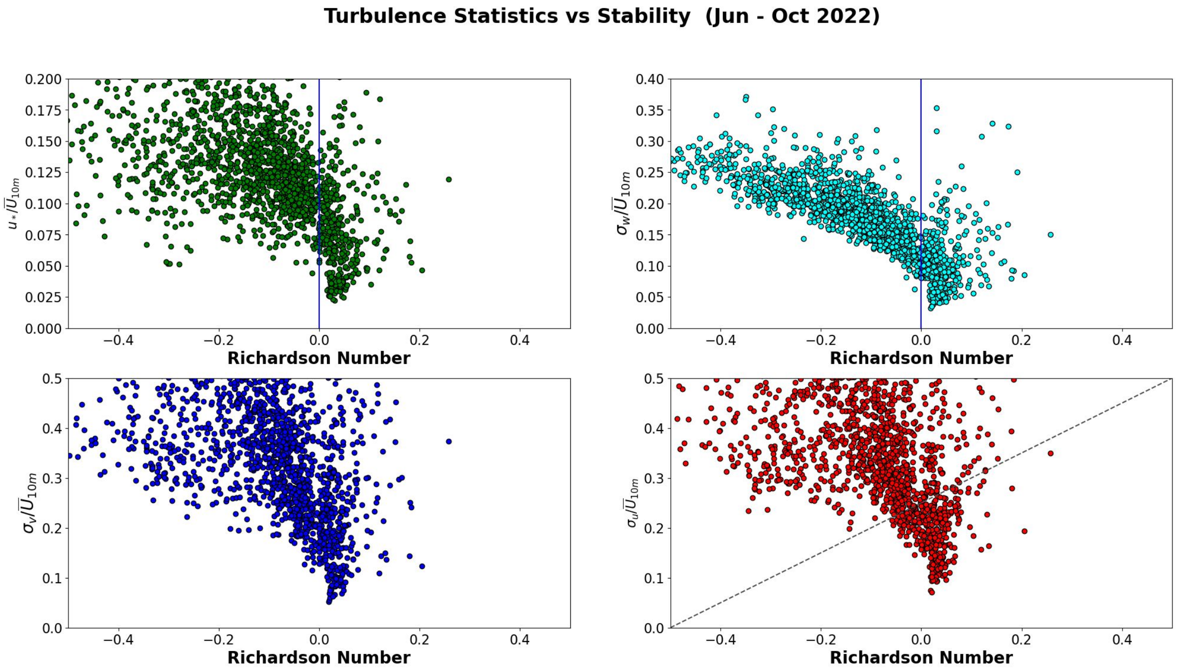
Eddy covariance at 3 and 10 m

LI-7500 DS for H2O and CO2

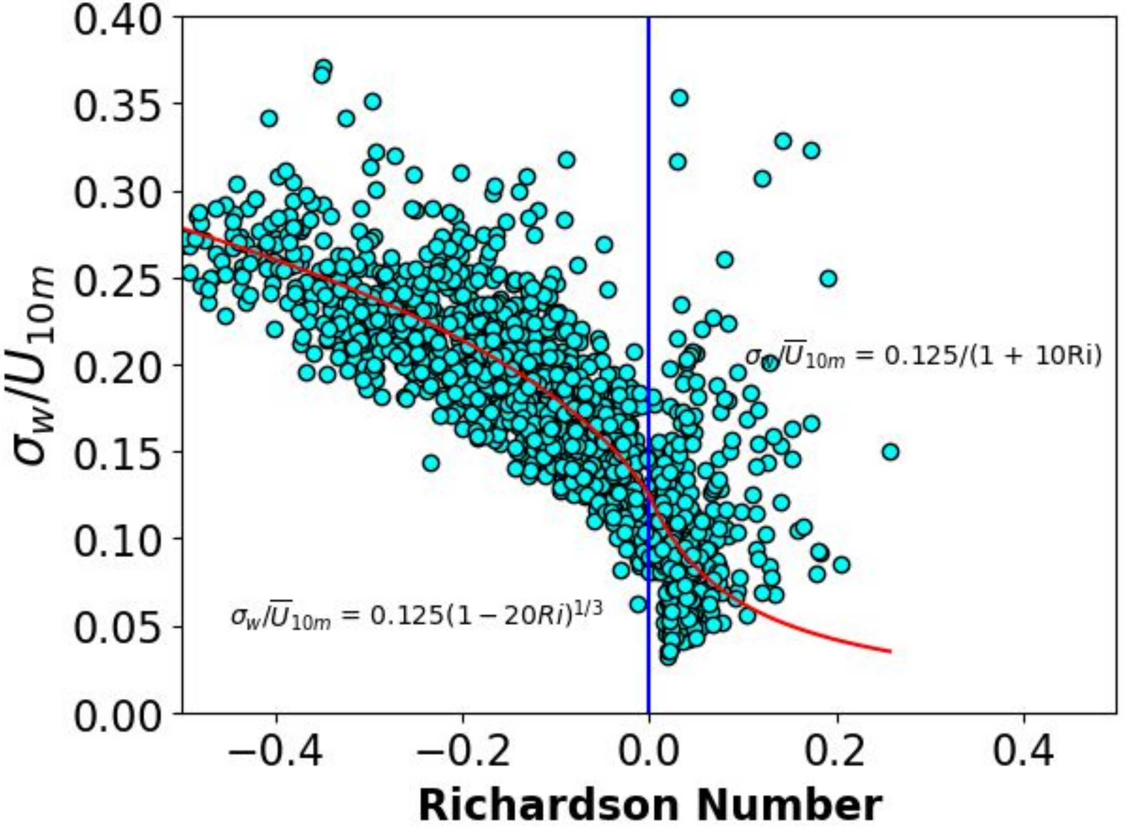
Smartflux system with RM Young VRE Sonic

EddyPro processing software

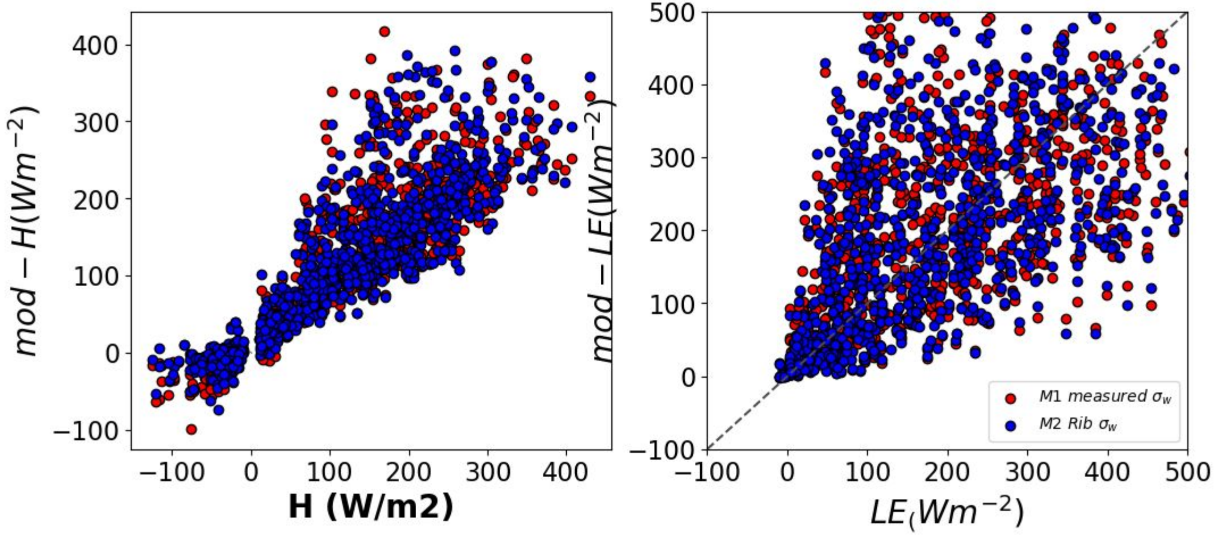
10 Hz sampling, output 30 minute average



- Complicated stable conditions
- Much more scatter for u,v, u*


0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.4
−0.2
0.0
0.2
0.4
Richardson Number
$\sigma_w/\overline{U}_{10m}$
$\sigma_w/\overline{U}_{10m} = 0.125/(1 + 10\mathrm{Ri})$
$\sigma_w/\overline{U}_{10m} = 0.125(1-20Ri)^{1/3}$

Simple Model Energy Fluxes- (Jun - Oct 2022)

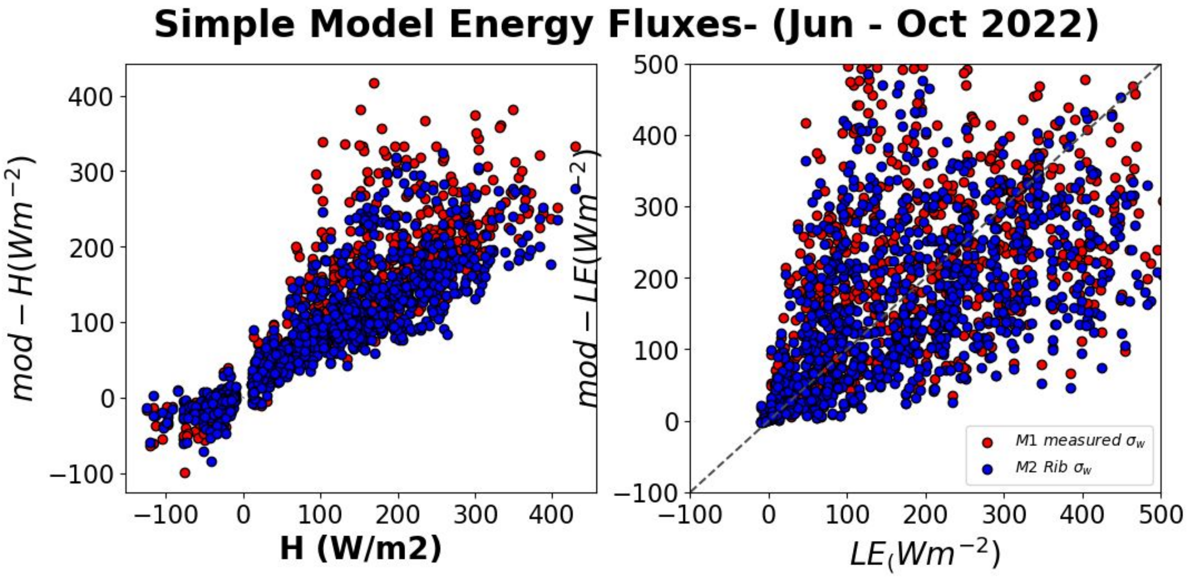


M1 M2

$$\overline{w'T'} = C\sigma_w(T_s - T_{air}) = CUf(Ri_b)(T_s - T_{air})$$

$$\overline{w'q'} = C\sigma_w(q_s - q_{air}) = CUf(Ri_b)(q_s - q_{air})$$

Using the BNV formulation



M1 M2

$$\overline{w'T'} = C\sigma_w(T_s - T_{air}) = CUf(Ri_b)(T_s - T_{air})$$

$$\overline{w'q'} = C\sigma_w(q_s - q_{air}) = CUf(Ri_b)(q_s - q_{air})$$

Conclusions

- σ_w/U can be written as a function of the bulk Richardson number, easier to measure routinely with vertical gradients and gap fill data
- Formulation from an ‘ideal’ site could be extended to more complex sites and fetches
- We still need some prior data from the particular field station to estimate the ‘bulk coefficient’
- Refine calculations for LE and q_s
- Investigate the dependence on Ts

Thank you!

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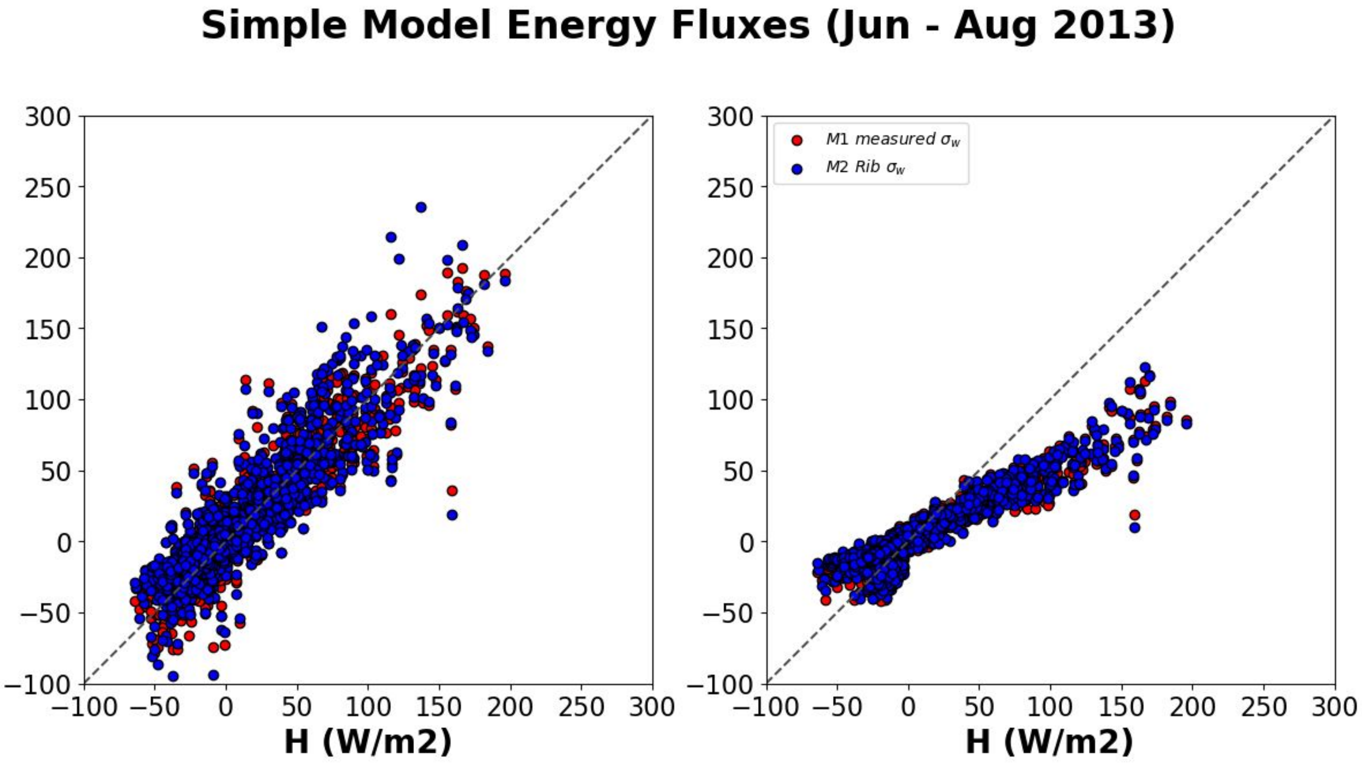
Fully parameterized fluxes BNV, comparing 2m air temp.



The modeled H is lower.
Need to capture the higher surface temps
How imp is it to use our improved method?
Will mostly move this to extra slides for now.

How well does this hold up for complex sites?

How imp. Is it to get the surface temperatures?